AI and Finance

Artificial Intelligence

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Contents

1 AI & Finance Summary

In the first part of the course, we will consider the calibration of parameters for stochastic models.

1.1 Data from the market

The idea: Given some data from the market

- past prices of stocks, bonds ... A bond is a piece of paper that promises to pay a certain amount of money at a certain date. You can buy a bond for a certain price and in the given date you will receive the money (considering an interest rate). Stocks are instead a share of a company. You have "no idea" of the future value of the company, you can only buy and sell them on the market. Prices contain information about the market.
- past interest rates. It changes over time and depends on market, company policy, etc. Interest rates are fundamental because they allow us to compare amounts of money at different points in time.
- option prices (call and put options). An option is a contract that gives the right to buy or sell a certain asset at a certain price at a certain date. Call option: gives the right to buy an asset at a certain price at a certain date. Put option: gives the right to sell an asset at a certain price at a certain date. It is a sort of insurance. This option are very liquid, meaning that they can be bought and sold very easily. Example: you want to buy steel in the future, but you think that the price will increase. You can buy a call option on steel. If the price of steel increases, you can exercise the option and buy steel at the price of the option. If the price of steel decreases, you can let the option expire and lose the money you paid for the option. Example: you want to sell steel in the future, but you think that the price will decrease. You can buy a put option on steel. If the price of steel decreases, you can exercise the option and sell steel at the price of the option.

These are all important types of data that we can use to calibrate parameters for stochastic models.

1.2 Stochastic Models

In finance there is a part of the description that is *deterministic*, but there is also a part that is *stochastic*. For example, we can consider the **Bernoulli model**.

1.2.1 Bernoulli Model

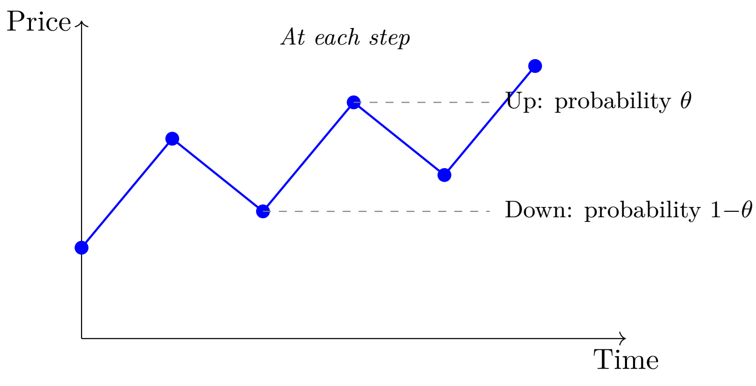

Figure 1: Bernoulli model: at each step, price can go up (probability θ) or down (probability $1-\theta$).

1.2.2 Black-Scholes Model

The **Black–Scholes model** is a stochastic model that describes the price of a stock over time. It is a continuous-time model used to price options. It is more complex than the Bernoulli model, but also more accurate. It is based on the assumption that the price of a stock follows a *geometric Brownian motion*. Ignoring the interest rate for the moment, we can write:

Black–Scholes dynamics (no drift)

In differential notation, the Black–Scholes model expresses the evolution of the asset price S as

$$dS_t = S_t \, \sigma_{BS} \, dW_t, \tag{1}$$

where:
- S_t is the asset price at time t,
- σ_{BS} is the volatility coefficient (a measure of how much the price can fluctuate),
- dW_t is an increment of a Wiener process (standard Brownian motion).

In a small time step δt, a discretised version can be written as

$$S_{t+\delta t} = S_t + S_t \, \sigma_{BS} \, (W_{t+\delta t} - W_t), \tag{2}$$

where $W_{t+\delta t} - W_t$ is again an increment of Brownian motion.

We basically add a variance proportional to the time step and to today's price. The parameter σ_{BS} is the volatility coefficient. A bigger σ_{BS} means that the price can fluctuate more, while if $\sigma_{BS} = 0$ the price is constant.

There are parameter-dependent (i.e. depending on σ_{BS}) prices of call and put options for each strike K and maturity T.

$$C^K, T(\sigma_B S, S_0) \quad \& \quad P^K, T(\sigma_B S, S_0) \tag{3}$$

1.2.3 Stochastic Volatility Model

We can improve the Black–Scholes model by introducing a *stochastic volatility*. We obtain a more complex model:

$$\begin{aligned} dS_t &= S_t \sqrt{V_t} \, dW'_t \\ dV_t &= \kappa(\theta - V_t) \, dt + \sigma_{SV} \sqrt{V_t} \, dW_t^{(2)} \\ d\langle W', W^{(2)} \rangle_t &= \rho \, dt \end{aligned} \tag{4}$$

where:

- W'_t and W_t^2 are two independent Wiener processes (standard Brownian motion).
- κ is the rate of mean reversion,
- θ is the long-term average volatility,
- $\sigma_S V$ is the volatility of the volatility.
- ρ is the correlation between the two Wiener processes.

The increments $(W'_{t+\delta t} - W'_t)$ and $(W^2_{t+\delta t} - W^2_t)$ are jointly normally distributed as

$$\mathcal{N}\left(0, \delta t \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right).$$

This leads to complicated parameter-dependent prices for call and put options. The goal is to find the best possible parameters for the model.

- Matching the observations on the market
- In a quick way

How can we do this? Typically with **regressions**, for example:

- classical approach: maximum likelihood estimation (MLE),
- Gaussian processes,
- neural networks.

1.3 Implied volatility and inverse pricing

What do we need to learn? We want to learn the mapping from option prices and, more generally, market data to the best possible parameters for the model. We are especially interested in approximating key functionals such as the *implied volatility*.

Pricing and implied-volatility maps

In the Black–Scholes model there is a pricing map

$$(\sigma_{BS}, S_0) \mapsto C^{K,T}(\sigma_{BS}, S_0), \tag{5}$$

and one can (in principle) construct a very complicated inverse map

$$C^{K,T}(\sigma_{BS}, S_0) \mapsto (\sigma_{BS}, S_0). \tag{6}$$

The *implied volatility* is the value of σ_{BS} such that the model price $C^{K,T}(\sigma_{BS}, S_0)$ matches the observed market price of the option.

If we now fix T:

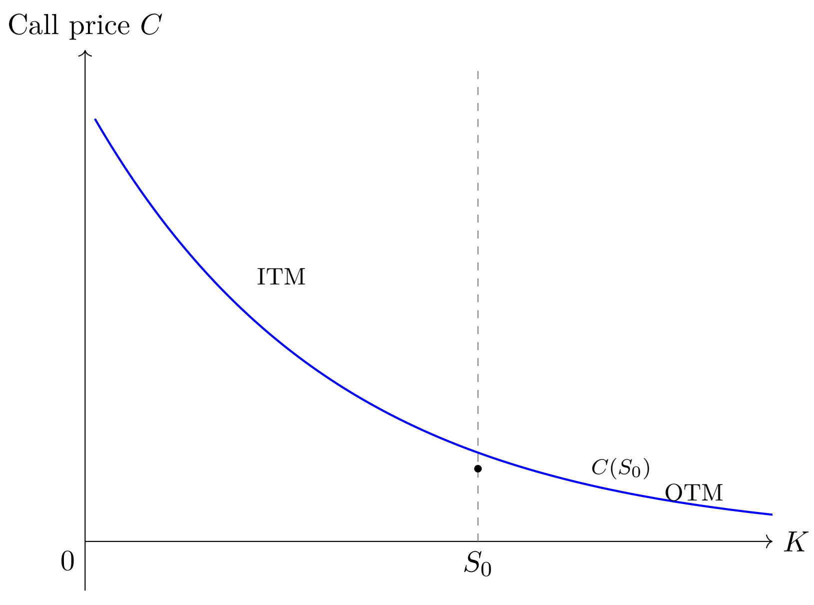

Figure 2: Schematic plot of the price C of a call option as a function of strike K, with S_0 located on the right half of the x-axis. The blue curve illustrates a decreasing, convex call-price profile.

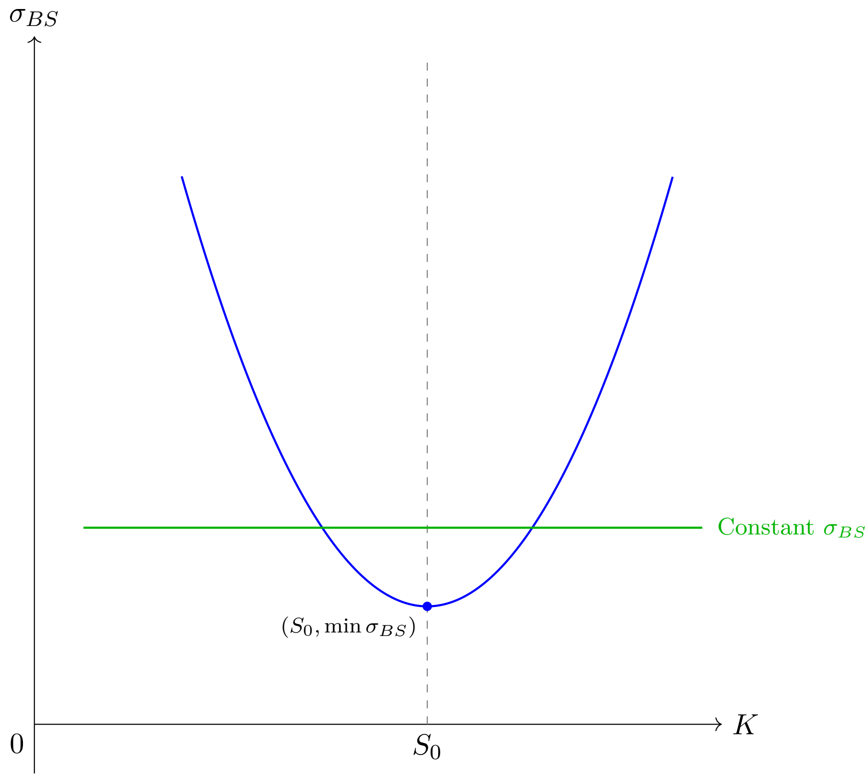

Figure 3: Schematic plot of implied volatility σ_{BS} as a function of strike K, with S_0 placed on the right half of the x-axis. The blue curve illustrates the volatility "smile" often observed in implied volatility, while the horizontal green line marks the constant Black–Scholes volatility.

Here we can apply the IV map and get the sigma of the Black-Scholes model. This is usually called the *volatility smile*. Often one prefers to replace prices with the implied volatility, since volatility is naturally scaled and still interpretable. More generally, we can view option prices as *pricing functionals* for complicated stochastic models or exotic options, mapping model parameters to theoretical prices.

Universal approximation theorem(s)

2 Time Series (second part of the course)

Given observations of available quantities over time:

- prices of stocks,
- bonds,
- options,
- interest rates
- historical values of variance and covariance
- historical values of traded volumes

Goal: **forecast** the next value of the corresponding quantity together with the corresponding *uncertainty*.

How: studying the time series (frequency, autocorrelation, stationarity, . . .)

1. Apply an econometric model:
   - ARMA(p, q) model:
     - $Y_t = \mu + \sum_{i=1}^{p} \phi_i Y_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} + \epsilon_t$
     - where:
       * Y_t: observation at time t (arrow above Y_{t-i}: "past observation")
       * ϵ_t: random shock/innovation at time t (arrow above both ϵ's: "shocks")
   - GARCH model:
     - calibration
     - evaluation of performance
2. Apply a RNN (recurrent neural network):
   - calibration (training)
   - evaluation of performance (testing)

2.1 Main applications

- Risk estimation of given portfolios (how can one measure risk?)
  - variance;
  - **Value at Risk (VaR)**: e.g. $\text{VaR}_{99\%} = 100$ if and only if the probability of having a loss ≥ 100 is approximately 1%;
  - **Expected Shortfall (ES)**: expected loss *conditional* on VaR being exceeded. This is typically called **CVaR** in engineering.

We use these models to forecast the distribution (mean, variance, . . .) of the value of a portfolio and use it to compute the corresponding risk. We then backtest the obtained predictions and perform portfolio optimization (e.g. *Markowitz* and *Reinforcement Learning* approaches).